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Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Evolutionary biology		
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Task Description:	The fundamental mechanism by which animal forms evolve is change in the genomic regulatory networks that control the developmental processes by which these forms are built in each life cycle: understanding evolution of animal forms comes down to understanding evolution of gene networks. But this is a yet unsolved problem because genomic regulatory networks are just beginning to be solved at the DNA level. We are in an unparalleled position to attack this problem because we have in hand what may be the first example of such a network, viz a network that underlies endomesoderm development in sea urchin embryos. We propose to perform a comparative analysis of network architecture in a starfish, the embryos of which appear to produce endomesoderm in a somewhat similar way. However, the last common ancestor of starfish and sea urchins lived >500 mya. A large amount of preliminary work has already been done: we have arrayed Asterina miniata (Am) BAC and embryo cDNA libraries; many endomesoderm regulatory		

	genes are already isolated and their expression characterized; and most importantly, we have developed an excellent method for gene transfer into Am eggs for study of gene regulation in development. Thus we are poised to carry out the first study of regulatory gene network evolution. Preliminary results suggest that many deeply informative surprises await us.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will help in understanding planetary bioscience.
Task Progress:	We have undertaken a comparative analysis of gene regulatory network (GRN) architectures governing endomesoderm development in the sea urchin Strongylocentrotus purpuratus and the sea star Asterina miniata. The gene regulatory network governing this process in the sea urchin is extremely well known. In previously reported work, we focused on elucidating particulars of the GRN for endomesodermal specification in the starfish in order to know which processes are conserved with the sea urchin (indicative an essential feature of their development), and alternatively how evolutionary differences are incorporated in GRNs. In the past years we focused more particularly on unraveling the cis-regulatory control region of the otx gene in A. miniata. The otx gene in both A. miniata and the sea urchin S. purpuratus is alternatively spliced to yield multiple transcripts and proteins. Studies in these echinoderms have focused on the otxß1/2 transcript which is zygotically activated and expressed in the ectoderm and vegetal plate of the late blastula embryo. This transcript is an important activator of many other transcription factors in the endomesoderm of both sea urchins and starfish. We have previously shown that in both animals, otxß1/2 is regulated by the same set of transcription factors, viz Krox, Gatae and Otx itself. These upstream factors are also similarly expressed in starfish and sea urchin. This conserved regulation of the otx gene in these two divergent echinoderms is part of series of a highly conserved feedback loop regulatory interactions that were presumably present in the ancestor of the two animals. Fossil records and ribosomal clocks indicate that such an ancestor lived in the upper Cambrian some 450-500 million years ago. The cis-regulatory control region of the sea urchin otx gene is well understood. There exist functional binding sites for the known regulatory proteins Krox, Gatae and Otx . In starfish, we isolated approximately 700bp of DNA downstream of the last otx exon that, when attached to any of several different basal promoters, drives reporter gene expression in the blastula and early gastrula stage embryo that mimics endogenous expression of the otxß1/2 transcript. This is therefore considered to be the region of DNA that normally regulates endogenous otxß1/2 in these stage starfish embryos. We have named this region of DNA OtxG. OtxG contains predicted consensus sequence for binding sites of Krox, Gatae and Otx proteins. Furthermore, this element contains consensus sequences for binding of the Tbrain factor. In recent work we show that in starfish, but not in sea urchin, the tbrain gene product is required for correct otx expression (in embryos in which the Tbrain protein is diminished, there is also diminished otxß expression). In starfish, tbrain is expressed similarly to the otxß transcript i.e. in the ectoderm and vegetal plate of the blastula, but is expressed earlier, beginning at around 12h after fertilization, c.f. 18h for the otxß transcript. In sea urchins, tbrain is expressed in the primary mesenchyme cell lineage and does not overlap with otxß expression). Therefore it is not expected for Tbrain to have a role in regulating otxß expression in the sea urchin. 1 We have undertaken extensive functional analysis of the OtxG reporter construct. When all three consensus Otx sites or both consensus Krox sites are mutated such that they should no longer bind the appropriate protein, there is diminished expression of the reporter. Similarly, mutating the combination of one otx site and its two adjacent gatae sites also results in diminished reporter expression. This shows, that just as for the corresponding element from the sea urchin, these consensus sites are functional and are necessary to bind the appropriate transcription factor to regulate endogenous otx expression. We then turned our attention to the role of Tbrain in regulating otxß in the starfish. When Tbrain protein is diminished in the embryo, OtxG reporter expression is also diminished. Also when all four putative tbrain binding sites are mutated such that they can no longer bind the Tbrain protein the reporter is also dramatically reduced. These two experiments are conclusive evidence that OtxG normally binds Tbrain protein to regulate otxß expression. When the sequence of OtxG and sea urchin otxß1/2 regulatory regions are compared using a variety of alignment algorithms no significant similarity can be found. Even the binding site sequences themselves are changed, although they still conform to the consensus sequence (which is degenerate). However we show that there is a similarity of the arrangements and types of binding sites and we also show that there is a functional conservation in the way in which these sites regulate the otx gene in the two animals. This is despite the 450-500 million years of independent evolution of these two animals. The most significant functional difference is the presence of four binding sites for the tbrain protein that are essential to drive expression in starfish. Although further work is required, it seems likely that functional sites for tbrain are not contained within sea urchin regulatory element. As a demonstration of this conservation we find that OtxG correctly drives reporter gene expression when injected into the sea urchin. This demonstrates that (i) the sea urchin regulatory proteins recognize binding sites contained within the regulatory DNA of starfish, and (ii) the arrangement of sites is appropriate to correctly integrate the upstream regulatory information (i.e. quantities of appropriate transcription factor proteins) to provide the correct output. Conversely when the sea urchin regulatory construct is injected into starfish only very reduced expression is observed. We hypothesis that this may be due to the lack of functional tbrain binding sites in the sea urchin regulatory region that are necessary to drive expression in the starfish. This work is being finished for publication. It will be an important demonstration of (i) the nature of the preservation of cis-regulatory information against a background of highly divergent sequence composition and (ii) how a single GRN change viz Tbrain regulation of otx, is effected by altering function binding sites in a cis-regulatory element.
Bibliography Type:	Description: (Last Updated: 07/18/2007)

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